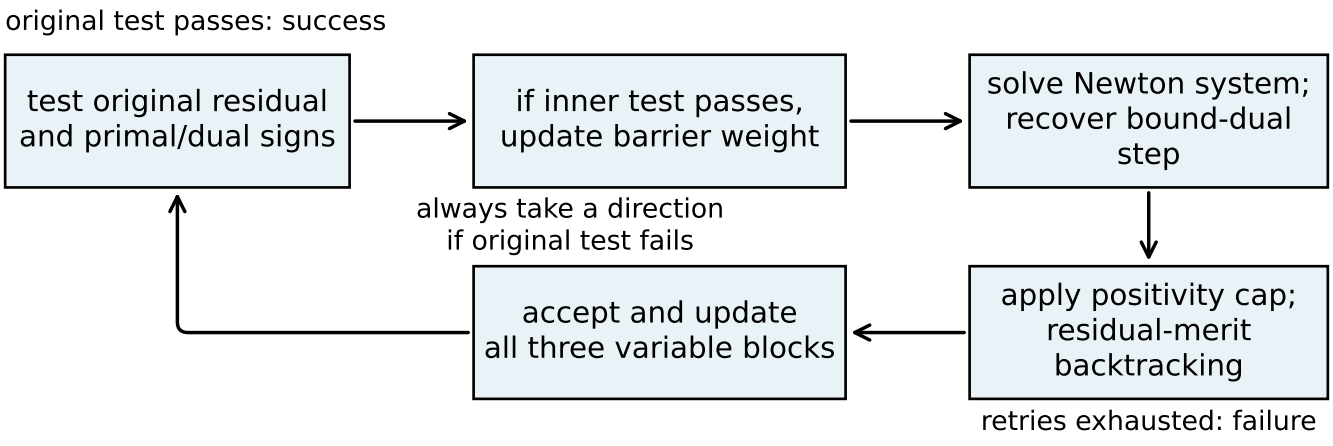
original test passes: success
test original residual and primal/dual signs
if inner test passes, update barrier weight
solve Newton system; recover bound-dual step
always take a direction if original test fails
accept and update all three variable blocks
apply positivity cap; residual-merit backtracking
retries exhausted: failure